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DEPARTMENT OF INFORMATION TECHNOLOGY

K.T. Marg, Vasai Road (W), Dist-Palghar - 401202, Maharashtra

Experiment no.11

Aim : To understand continuous monitoring and installation and Configuration of Nagios

Theory :

1. Introduction to Continuous Monitoring

Continuous monitoring refers to the ongoing process of observing the health, performance, and availability of IT infrastructure, applications, and services in real-time. It allows organizations to proactively detect problems, prevent downtime, and ensure system reliability.

Key aspects of continuous monitoring include:

- **Performance Monitoring:** Tracking CPU, memory, disk usage, network latency, etc.
- **Availability Monitoring:** Ensuring servers, applications, and services are online and accessible.
- **Alerting and Notifications:** Immediate alerts when thresholds are breached or failures occur.
- **Reporting and Analysis:** Historical data helps in capacity planning and troubleshooting.

Benefits:

1. Reduces downtime and improves system reliability.
 2. Enables proactive problem detection and resolution.
 3. Helps in maintaining SLAs (Service Level Agreements).
 4. Supports capacity planning and resource optimization.
-

2. Introduction to Nagios

Nagios is a popular open-source monitoring tool used for continuous monitoring of systems, networks, and infrastructure. It helps detect problems, alert administrators, and ensure IT services are running smoothly.

Key Features of Nagios:



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- **Monitoring Capabilities:** Monitors servers, switches, applications, databases, and network protocols.
- **Alerting System:** Sends alerts via email, SMS, or custom scripts when issues arise.
- **Visualization:** Provides web-based dashboards showing real-time status and trends.
- **Plugin Support:** Highly extensible using community or custom plugins for specific monitoring needs.
- **Reporting:** Generates reports for uptime, performance trends, and SLA compliance.

Nagios Architecture:

1. **Nagios Core:** The central monitoring engine that schedules checks, processes results, and triggers alerts.
 2. **Plugins:** Scripts or binaries that perform actual checks (e.g., check CPU load, check disk space).
 3. **NRPE (Nagios Remote Plugin Executor):** Enables monitoring of remote Linux/Unix systems.
 4. **Web Interface:** Provides a graphical view of hosts, services, and alerts.
-

3. Installation of Nagios

Nagios can be installed on Linux servers (commonly Ubuntu or CentOS). Steps usually include:

1. Install required dependencies (Apache, PHP, GCC, GD library).
 2. Download Nagios Core source files.
 3. Compile and install Nagios.
 4. Install and configure Nagios Plugins.
 5. Start the Nagios service and enable it at boot.
 6. Access the Nagios web interface via browser.
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4. Configuration of Nagios

After installation, Nagios must be configured to monitor hosts and services. Key configuration steps include:

- 1. Defining Hosts:** Specify IP addresses or hostnames of systems to monitor.
 - 2. Defining Services:** Specify what to monitor on each host (e.g., HTTP service, disk usage, CPU load).
 - 3. Defining Contacts:** Specify who receives alerts.
 - 4. Defining Notification Rules:** Set thresholds and conditions for alerts.
 - 5. Testing and Reloading Configuration:** Ensure Nagios correctly monitors all specified hosts and services.
-

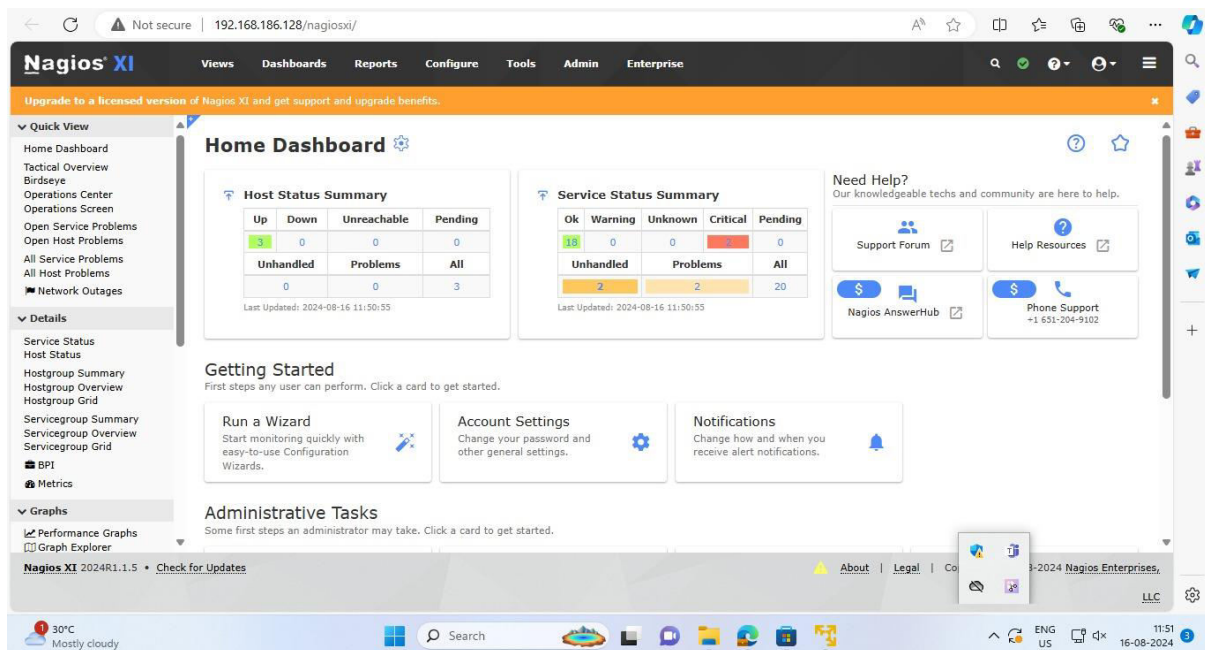
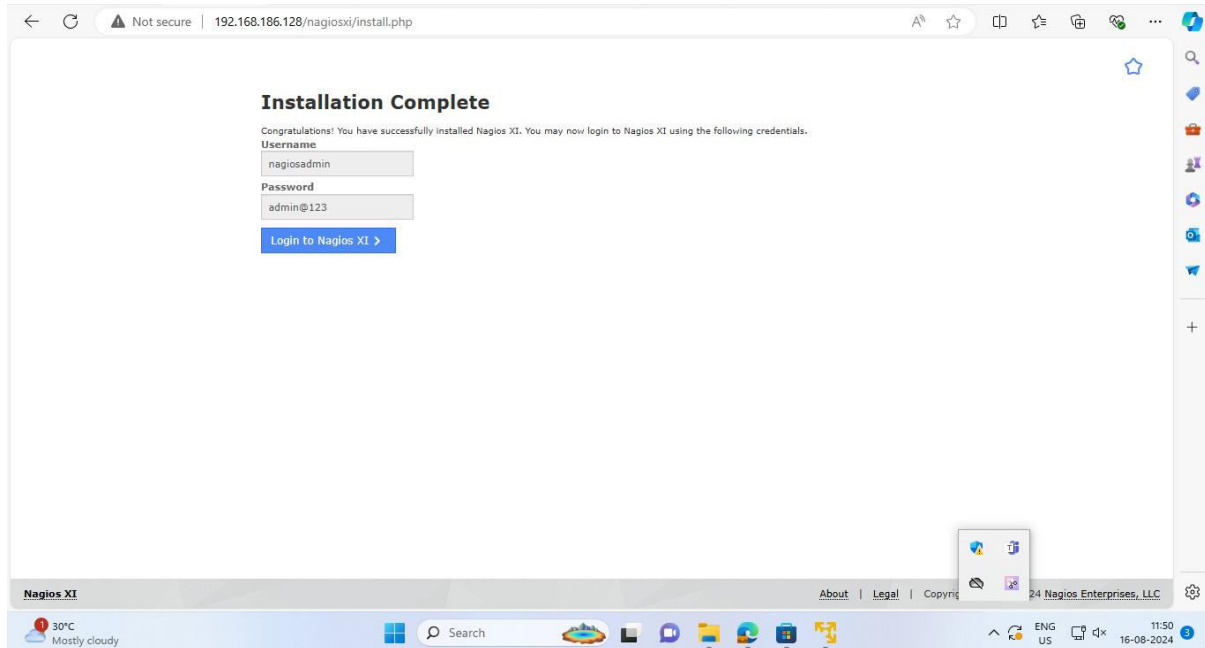
5. Benefits of Using Nagios for Continuous Monitoring

- 1. Proactive Problem Detection:** Alerts administrators before small issues become critical.
- 2. Centralized Monitoring:** Monitor multiple hosts and services from a single dashboard.
- 3. Flexibility and Extensibility:** Supports custom plugins and scripts for tailored monitoring.
- 4. Improved Reliability and Uptime:** Ensures IT systems are always operational.
- 5. Historical Reporting:** Tracks trends over time to optimize resources and plan upgrades.



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Procedure :





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Configuration Options

Choose what how you would like to configure Nagios XI. To get started right away, try using a configuration wizard under the 'Start Monitoring Now' section.

Deploy Monitoring Agents and Configure

Quickly deploy NCPA (our monitoring agent) on a Linux system and configure what you'd like to monitor.

[Run Auto Deployment >](#)

Start Monitoring Now

Quickly monitor a new device, server, application, or service using an easy configuration wizard.

[Run a configuration wizard >](#)

Auto-Discovery

Run an auto-discovery job to automatically find hardware, devices, and services to monitor.

[Use the auto-discovery tool >](#)

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Website Configuration Wizard Step 1

Website Information

Website URL [ⓘ](#)

[Next >](#) [Cancel](#)

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Service Status

Host: www.geeksforgeeks.org

Host Status Summary

Up	Down	Unreachable	Pending
1	0	0	0
Unhandled	Problems	All	
0	0	1	

Last Updated: 2024-08-16 11:53:41

Service Status Summary

Ok	Warning	Unknown	Critical	Pending
3	0	0	1	0
Unhandled	Problems	All		
1	1	4		

Last Updated: 2024-08-16 11:53:41

Showing 1-4 of 4 total records

Page 1 of 1 15 Per Page Go Search...

Host	Service	Status	Duration	Attempt	Last Check	Status Information
www.geeksforgeeks.org	DNS IP Match	Critical	26m 27s	5/5	2024-08-16 11:52:14	DNS CRITICAL - expected '18.172.78.124' but got '18.172.78.124,18.172.78.26,18.172.78.45,18.172.78.53'
	DNS Resolution	Ok	29m 38s	1/5	2024-08-16 11:49:03	DNS OK: 0.021 seconds response time. www.geeksforgeeks.org returns 18.172.78.124,18.172.78.26,18.172.78.45,18.172.78.53
	HTTP	Ok	28m 57s	1/5	2024-08-16 11:49:44	HTTP OK: HTTP/1.1 301 Moved Permanently - 566 bytes in 0.012 second response time
	Ping	Ok	28m 9s	1/5	2024-08-16 11:50:32	OK - www.geeksforgeeks.org rta 1.827ms lost 0%

Last Updated: 2024-08-16 11:53:41

Page 1 of 1 15 Per Page Go

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Configure

- Configuration Options
- Configuration Tools
 - Configuration Wizards
 - Auto-Discovery
 - Manage Templates
 - Top / Recent Wizards
- Auto Deployment
 - Deploy Agent
 - Past Deploy Agent Jobs
 - Manage Deployed Agents
 - Deployment Settings
- Advanced Configuration
 - Core Config Manager
- More Options
 - My Account Settings
 - System Configuration
 - User Management
 - Unconfigured Objects
 - Deadpool Settings

</> XML Configuration Wizard Step 1

Server Information

* Server Address

www.geeksforgeeks.org

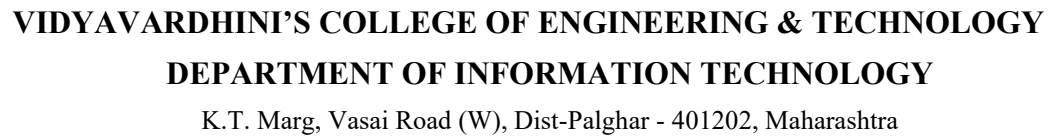
Server Port

80

Next > Cancel

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The screenshot displays the Nagios XI XML Configuration Wizard, Step 3: Monitoring Settings. The interface is clean and modern, with a dark header bar containing the Nagios XI logo and navigation links. The main content area is white with a blue sidebar on the left. The sidebar lists various configuration options, including 'Configure', 'Configuration Options', 'Configuration Tools', 'Auto Deployment', 'Advanced Configuration', and 'More Options'. The main content area is titled 'XML Configuration Wizard' and 'Step 3'. The 'Monitoring Settings' section defines parameters for how the host and service(s) should be monitored. Under normal circumstances, the host and service(s) are monitored every 5 minutes. When a potential problem is first detected, the host and service(s) are re-checked every 1 minute, up to 5 times before sending a notification. The wizard includes a sidebar with navigation options like 'Configure', 'Configuration Options', 'Configuration Tools', 'Auto Deployment', 'Advanced Configuration', and 'More Options'. The bottom of the screen shows the Nagios XI version (2024R1.1.5) and a 'Check for Updates' link.

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Not secure | 192.168.186.128/nagiosxi/config/

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</> XML Configuration Wizard

✓ Configuration applied successfully.

- Your configuration changes have been successfully applied and the monitoring engine was restarted.

Configuration Request Successful

[Run this monitoring wizard again](#) [Run another monitoring wizard](#)

Other Options:

- [View the latest configuration snapshots](#)

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Configuration Snapshots

The latest configuration snapshots of the monitoring engine are shown below. Download the most recent snapshots as backups or get vital information for troubleshooting configuration errors.

Snapshots

Date Created	Config Status	Snapshot Name	Diff Changes	Actions
2024-08-16 16:31:03	OK	Snapshot 1723806063	N/A	Download Refresh Delete
2024-08-16 11:58:12	OK	Snapshot 1723789692	-2 +250	Download Refresh Delete
2024-08-16 11:53:20	OK	Snapshot 1723789400	-2 +250	Download Refresh Delete
2024-08-16 11:37:49	OK	Snapshot 1723788469	-2 +250	Download Refresh Delete
2024-08-16 11:24:11	OK	Snapshot 1723787651	-2 +250	Download Refresh Delete
2024-08-16 11:22:32	OK	Snapshot 1723787552	-2 +250	Download Refresh Delete
2024-08-16 11:19:13	OK	Snapshot 1723787353	-2 +126	Download Refresh Delete
2024-07-22 20:39:02	OK	Snapshot 1721660942	N/A	Download Refresh Delete
2024-07-22 20:38:46	OK	Snapshot 1721660926	N/A	Download Refresh Delete

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Configuration Snapshots - Config Changes

Changes in configuration between the **current running configuration** (on the left) and the snapshot at **2024-08-16 16:31:03** (on the right) are shown below.

File	Current Running Configuration	Snapshot (2024-08-16 16:31:03)
usr/local/nagios/etc → nagiosxi/nom/checkpoints/nagioscore/1723806063/contacts.cfg	<pre>@@ -3,7 +3,7 @@ 3 # Contacts configuration file 4 # 5 # Created by: Nagios CCM 3.1.9 6 - # Date: 2024-08-16 11:58:11 7 # Version: Nagios Core 4.x 8 # --- DO NOT EDIT THIS FILE BY HAND --- 9 @@ -22,7 +22,7 @@ define contact { 22 service_notification_period nagiosadmin_notification_t 23 host_notification_options d,u,r,f,s 24 service_notification_options w,u,c,r,f,s 25 - email root@localhost 26 use xi_contact_generic 27 } 28</pre>	<pre>3 # Contacts configuration file 4 # 5 # Created by: Nagios CCM 3.1.9 6 + # Date: 2024-07-22 10:08:45 7 # Version: Nagios Core 4.x 8 # --- DO NOT EDIT THIS FILE BY HAND --- 9 @@ -22,7 +22,7 @@ define contact { 22 service_notification_period nagiosadmin_notification_t 23 host_notification_options d,u,r,f,s 24 service_notification_options w,u,c,r,f,s 25 + email nagios@localhost 26 use xi_contact_generic 27 } 28</pre>
usr/local/nagios/etc/hosts/ashishvanmali.wordpress.com.cfg	DELETED	

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Nagios XI Server Configuration Wizard

Step 1

Nagios XI Server
Specify the details for the remote Nagios XI server you want to monitor.

* Address

* URL

Authentication Settings
An admin user's API key for the Nagios XI server is required.

☐ Legacy Nagios XI authentication

* API Key

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Upgrade to a licensed version of Nagios XI and get support and upgrade benefits.

▼ Configure
■ Configuration Options
▼ Configuration Tools
✔ Configuration Wizards
🔍 Auto-Discovery
📁 Manage Templates
📁 Top / Recent Wizards
▼ Auto Deployment
▶ Deploy Agent
▶ Past Deploy Agent Jobs
📁 Manage Deployed Agents
⚙️ Deployment Settings
▼ Advanced Configuration
⚙️ Core Config Manager
▼ More Options
🔍 My Account Settings
🔍 System Configuration
🔍 User Management
🔍 Unconfigured Objects
🔍 Deadpool Settings

XI Nagios XI Server Configuration Wizard

 Step 2 ⚙️ ⓘ

Nagios XI Server

Address

192.168.186.128

Host Name ⓘ

localhost.localdomain

URL

http://192.168.186.128/nagiosxi/config/

Nagios XI Server Metrics

Specify the metrics you would like to monitor on the remote Nagios server.

☒ Ping ⓘ

☒ Nagios XI Web Interface ⓘ

☒ Monitoring Daemons ⓘ

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▼ Configure
■ Configuration Options
▼ Configuration Tools
✔ Configuration Wizards
🔍 Auto-Discovery
📁 Manage Templates
📁 Top / Recent Wizards
▼ Auto Deployment
▶ Deploy Agent
▶ Past Deploy Agent Jobs
📁 Manage Deployed Agents
⚙️ Deployment Settings
▼ Advanced Configuration
⚙️ Core Config Manager
▼ More Options
🔍 My Account Settings
🔍 System Configuration
🔍 User Management
🔍 Unconfigured Objects
🔍 Deadpool Settings

XI Nagios XI Server Configuration Wizard

 Step 3 ⚙️ ⓘ

Monitoring Settings

Define basic parameters that determine how the host and service(s) should be monitored.

Under normal circumstances:

Monitor the host and service(s) every minutes

When a potential problem is first detected:

Re-check the host and service(s) every minutes up to times before sending a notification

< Back

Next >

Finish with Defaults

Cancel

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Configure

- Configuration Options
- Configuration Tools
 - Configuration Wizards
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 - Core Config Manager
- More Options
 - My Account Settings
 - System Configuration
 - User Management
 - Unconfigured Objects
 - Deadpool Settings

XI Nagios XI Server Configuration Wizard

✓ Configuration applied successfully.

- Your configuration changes have been successfully applied and the monitoring engine was restarted.

Configuration Request Successful

[Run this monitoring wizard again](#) [Run another monitoring wizard](#)

Other Options:

- [View status details for localhost.localdomain](#)
- [View the latest configuration snapshots](#)

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High UV Now

← ↻ ⚠ Not secure | 192.168.186.128/nagiosxi/includes/components/xicore/status.php?host=localhost.localdomain

Service Status

Host: localhost.localdomain

Host Status Summary

Up	Down	Unreachable	Pending
1	0	0	0
Unhandled	Problems	All	
0	0	1	

Last Updated: 2024-08-16 12:06:54

Service Status Summary

Ok	Warning	Unknown	Critical	Pending
2	0	0	3	0
Unhandled	Problems	All		
4	4	6		

Last Updated: 2024-08-16 12:06:54

Showing 1-6 of 6 total records

Page 1 of 1 15 Per Page Go

Host	Service	Status	Duration	Attempt	Last Check	Status Information
localhost.localdomain	HTTP	Ok	N/A	1/5	2024-08-16 12:02:23	HTTP OK: HTTP/1.1 200 OK - 2838 bytes in 0.001 second response time
	I/O Wait	Critical	11s	5/5	2024-08-16 12:06:43	Error: Could not parse JSON from http://192.168.186.128/nagiosxi/config/ ()
	Load	Critical	2m 44s	3/5	2024-08-16 12:06:09	Error: Could not parse JSON from http://192.168.186.128/nagiosxi/config/ ()
	Nagios XI Daemons	Critical	2m 25s	3/5	2024-08-16 12:06:32	Error: Could not parse JSON from http://192.168.186.128/nagiosxi/config/ ()
	Nagios XI Jobs	Critical	1m 31s	2/5	2024-08-16 12:06:26	Error: Could not parse JSON from http://192.168.186.128/nagiosxi/config/ ()
	Ping	Ok	N/A	1/5	2024-08-16 12:06:15	OK - 192.168.186.128 rta 0.015ms lost 0%

Last Updated: 2024-08-16 12:06:54

Page 1 of 1 15 Per Page Go

30°C Mostly cloudy



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Conclusion :

Continuous monitoring is essential in modern IT environments to maintain reliability, prevent downtime, and improve performance. Nagios is a powerful tool that enables such monitoring by providing real-time visibility, alerting, and reporting. Understanding its installation and configuration allows IT administrators to maintain robust, highly available infrastructure.